

Module 11: Capital Structure

Topic 200: Capital Structure-Introduction

Capital Structure:

In simple words, **Capital Structure** shows that what is the percentage of debt and equity in any business or company. It means that **Capital Structure** actually comprises of debt (I.e., loan, bonds etc.) or equity (I.e., common stocks, preferred stocks) or a combination of these three things is called Capital Structure.

Mix of Debt and Equity: If the percentage of equity becomes increased and percentage of debt or preferred stock also varies then we can say that Capital Structure gets changed. So, a given **Capital Structure** simply means that there is a fixed percentage of debt and equity in this business.

Financial Leverage: With reference to Capital Structure, the interesting concept which actually affects the Capital Structure is Financial Leverage. The **Financial Leverage** simply determines the percentage of debt in the total capital meaning that if we have capital of 100 million then how much is the debt in that capital. Let's say if the debt is 40 million so, the leverage is 40%.

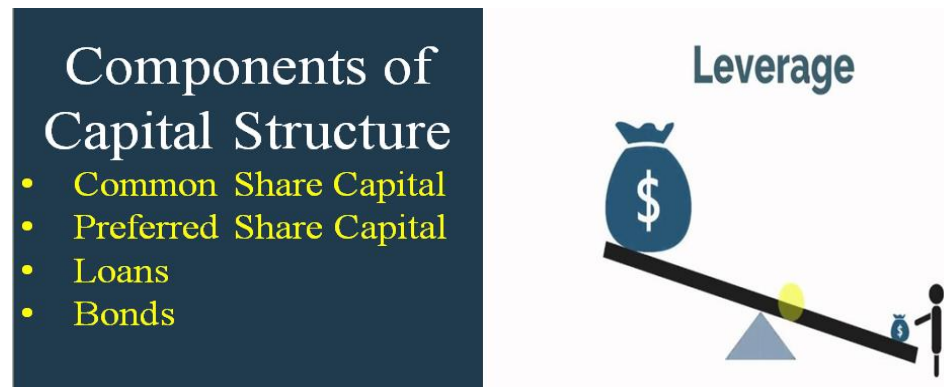
The basic thing that we need to understand here is that as the financial leverage get increase, the return of shareholder or equity holders also get increased. So, when we are changing the Capital Structure by increasing the financial leverage and reducing the percentage of equity, the return to the equity holders increases or we can say that it should increase.

What does this financial leverage do?

When we borrow money through banks or bonds or take debt then it creates Financial Leverage. Suppose we have borrowed money as a business or company at the rate of 10% and invested that borrowed amount in our business and have earned 15% profit or return per year from that borrowed amount and after giving the 10% to the bond holders/investors/loan providers from the earned 15%

return or profit, we have left with 5% additional profit which will be added into our shareholders or equity holders' profit.

The return of shareholders or equity holders will not be 15% but it will increase to 20% and this is the impact of Financial Leverage.



This picture also depicts that with the lesser amount of investment, we get greater return and this is the job of lever meaning that **lever** is such an invention through which we can put up more weight by applying less force. So, **Financial Leverage** actually refers to that with a smaller amount that we get higher return.

Components of Capital Structure:

Common share capital: This is one part of the Capital Structure in which the financial leverage would affect as a positive return.

Preferred share capital: Then there are preferred shareholders, they are not directly affected by financial leverage generated to the bonds but one view point is that preferred shares because of the fixed cost also create a financial leverage in a company.

Loan: This is also part of Capital Structure.

Bonds: This is also part of Capital Structure.

So, we can say that in this Capital Structure (in which there are common shares, preferred shares, public and private debts etc.), and financial leverage creates from where.

We can conclude our discussion that **Financial Leverage** mainly is generated by loan, bonds and other debts but the fixed cost of preferred share capital also creates a financial leverage. And when we say that there happening changes in the Capital Structure then actually the percentage of financial leverage is increasing or decreasing and mainly this leverage is due to the debt.

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Topic 201: Goals of a Firm

In this lecture, we will talk about the goals or the objects of a firm or a business or a company or any entity which is a commercial entity.

Profit Maximization:

The basic objective is certainly Profit Maximization. Every business that is established for the reason to earn profit from it, it is a commercial business then its prime objective is to maximize the profit and the term that we use is **Profit Maximization**. So, the prime objective of any business from the perspective of management is that to maximize its profit.

Maximization of Wealth:

If we have started our business with Rs.100 and earn profit of Rs.10 after one year then our profit of Rs.10 will be added in our wealth and in this way, our overall wealth becomes from Rs.100 to Rs.110. So, looking at the concept of Maximization of Profit and Maximization of Wealth seems to be very similar but the slight difference is that when we talk about the Maximization of Wealth, it is an absolute term meaning that how much amount of investment has been increased in rupees or dollars and we try to increase more and more in commercial ventures and that is called **Maximization of Wealth**.

Maximization of profit means that how much maximum percentage of profit we want to earn from our investment and we can say that if there will be more profit

then there will be increase in wealth also but in some situations there can be some difference in it.

But **Maximization of Wealth or Maximization of Profit** is the prime objective of every business organization or commercial entity.

Other Goals of a Firm:

In addition to this basic objective or goal then there are some other goals which are given below:

CSR (Corporate Social Responsibility): We say that there is a responsibility on every commercial organization that in which society or country or economy it is operating then there is a need to take care of its environment, provide facilities for their people and don't create difficulties for them so, it is the social responsibility of any business organization or corporation. So, one of the goals may be to fulfill the Corporate Social Responsibility (CSR).

Economic Uplift: Every commercial organization or a business organization also contributes to the economy, as the business will expand because it is the part of any economy then this business will also expand economy. So, an objective is also to uplift the economy of the country where the business is operating.

Employment Generation: It means that it is also objective to provide employment to more people.

We can conclude our discussion that we can say that **prime objective** of every business commercial organization is to maximize wealth and to maximize profit.

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Topic 202: Value of Firm: Capitalization Rate

Value of Firm:

Value of a firm means that the value which is there in the market.

Market value: Value of a firm may be its market value meaning that total value of shares or bonds of any company in the stock market determine the **Market Value of that firm** or tell us that this is the value of these shares in the market. And this is actually value that we need to consider while going through all discussion of Capital Structure and also in the next lectures.

Book Value: Another value is the Book value of the firm meaning that according to our Balance Sheet, what is the size of our firm. **Balance sheet** shows our financial position whose on one side there are assets and on other side there are debts and equity. In our books or financial statements, to be very precise in the Balance Sheet, how much is the value of that business and this is called the **Book Value** of that firm.

Other Value: Then there can be other values that may be a going concern value or can be intrinsic value and many other value that we can determine but our discussion will focus on the market value of the firm.

We use a concept to determine the market value of firm that we call

Capitalization. The **Capitalization** means that to show any expense as an asset or to calculate a value by dividing any amount by rate of return or percentage.

So, when we are Capitalizing something that means we are dividing a certain amount with a certain rate and the result is a certain value meaning that if we divide the amount of dividend or amount of return received on any equity or common share by its percentage cost then we will get the value of share. And if we divide the interest of any bond by its percentage return then we will get the value of that bond.

So, capitalizing the value of a bond or an equity or overall firm is done by dividing it with a certain percentage.

Now this is done by using a Capitalization Rate. The **Capitalization Rate** is used to capitalize certain return or a certain amount of profit or interest. It means that when we are going to capitalize any amount and that amount may be the dividend or interest also then we divide it by a percentage rate and that rate is called **Capitalization Rate.**

So, we will have an Equity Capitalization Rate and we may have a Bond or Interest Capitalization Rate and we may also have the cost of overall capital of the firm or company which may be called the **overall Capitalization Rate.**

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Topic 203: Capital Structure Theories

Before we talk about the Capital Structure Theories, let's first see that what is a Theory.

There is a misconception about it that Theory is just an assumption or it is just something that is very bookish and it does not have any practical implication. A **Theory** is basically a concept or an idea which is when to be communicated with others is converted into a statement that is called a proposition.

For example, a proposition may be that Capital Structure has a direct impact on the value of a firm. Now, this proposition has to be proved by experimentation or by survey or by some other method. So, this will be tested at different points of times, at different places with different people and if there is a consistency in the results we say that this proposition is accepted and it is converted into a Theory.

Theories related to Capital Structure:

There are mainly three theories in the Capital Structure:

- 1) Traditional Theory
- 2) Net Operating Income Theory
- 3) M & M Theory or Miller- Modigliani's argument on Capital Structure Theory

Traditional Theory:

This theory simply states that the value of a firm is dependent on the Capital Structure that is; if we change the Capital Structure then the value of firm will change and if the Capital Structure is altered by increasing or decreasing the leverage that affects the overall cost of capital.

Net Operating Income Theory:

It is a contrast to the Traditional Theory of Capital Structure and this theory states that the value of firm is not dependent on the Capital Structure. So, whatever is the Capital Structure, the value of the firm will not change.

M & M Theory or Miller- Modigliani's argument on Capital Structure Theory:

Miller- Modigliani's argument is actually in favor of Net Operating Income Theory.

The Traditional Theory is called the Theory of Relevance and the other two (Net Operating Income Theory and M & M theory) are called the theories of Irrelevance.

Let's have a look at some of the important concepts in these theories which will be used in our discussion.

Cost of Debt: The first one is cost of debt, it is the interest or the financial cost that we have to bear as a company on the amount of debt that we borrow. So, if we

take loan or issue any bond then the cost or interest that we have to pay on that loan or bond which is the cost of debt.

Cost of Equity: It is the profit that we earn for the shareholders which is paid in the form of dividend. So, the cost of Equity is the return to the shareholder.

Overall cost of capital: The overall cost of capital can be the weighted average cost of capital or they may be some other method to calculate it but this is the cost which is used to calculate the overall value of a firm.

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Topic 204: Optimal Capital Structure

We will have a discussion on the Optimal Capital Structure. The term **Optimal** means maximum or something that cannot be improved or increased.

When we talk about **Capital Structure Theories**, our objective is basically to see whether Capital Structure is in any way linked to the value of a firm and if it does then what is that Capital Structure where the value of a firm becomes maximum.

The Capital Structure where the value of a firm becomes maximum that is actually called the **Optimal Capital Structure.**

Is there any Optimal Capital Structure?

This is basically the difference between the two point of views which are called the Relevance Theories and Irrelevance Theories of Capital Structure.

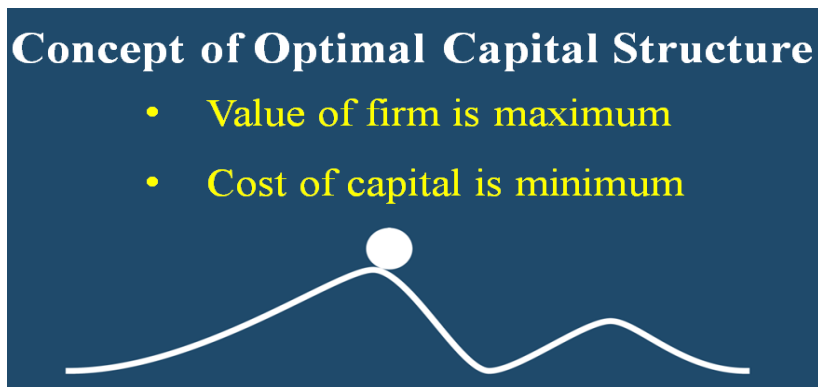
The first theory which is called the **Traditional Theory of Capital Structure** assumes that there exist a Capital Structure which is Optimal meaning that there is such a point of comparison of debt and equity where the value of firm becomes maximum.

How to calculate the value of a firm?

A firm which will earn profit or return and that return will continue forever till infinity and when we discount the values that will come in the form of dividend, Cashflows or returns from which we determine the value of a firm.

If there is an overall cost of capital of a firm or the weighted average cost of capital of the firm and if it is used to discount the profit or the earnings of the company, we get the value of firm.

So, value of firm is that to divide the profit with the overall cost of Capital. Now, if this overall cost of capital is somehow affected by the Capital Structure then the value of firm will change. So, the **Maximization of the value of a firm** means that how can we reduce the overall cost of capital by altering the Capital Structure meaning that we continue to change the proportion of our debt and equity and if we have 20% debt in start then we increase it as 30% or 45% and for each that option, we have to calculate our overall cost of capital then there becomes such a point (as Traditional Theory says) where our cost becomes minimum and when we divide this minimum cost with our amount of profit then value of firm becomes maximum.



So, when we say that there exists an Optimal Capital Structure then we can say that by changing the Capital Structure, the cost of capital changes. And when the cost of capital changes, it reduces and then it increases. When this cost becomes minimum then value of firm becomes maximum and we call that point an **Optimal Capital Structure**.

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Topic 205: Traditional Theory of Capital Structure-1

We are starting our discussion on the three basic theories of Capital Structure.

The **Traditional Theory of Capital Structure** in simple words states that there exist an Optimal Capital Structure so, we need to look for that best Capital Structure where our value of firm will be maximum.

Leverage and Required Rate:

Now, a Capital Structure is a function of Leverage and **Leverage** is actually the percentage of debt in the Capital Structure. The Capital Structure is basically the name of two things debt and equity and third thing that we call preferred shares capital also becomes the part of our Capital Structure. But if any firm increases the proportion of its debt then leverage will increase.

A basic thing that we need to keep in mind that leverage increases the return of shareholders or equity holders that means if there is more leverage meaning that if the percentage or proportion of debt will be more then the return of shareholders or equity holders will get increased.

Now, this is something which the equity holders or shareholders also know that whenever a company changes its Capital Structure by increasing the percentage or proportion of leverage in the total Capital Structure, the return of the equity holders is likely to increase. So, in anticipation the equity holders increase their required rate of return.

How do it works?

Earning Capacity: It states that a business has how much capacity to earn profit or return from the amount of investment meaning that some businesses have more capacity to earn more profit and some have less. So, this is the return on the investment and that investment may be in the form of both debt or equity. So, first we will see that what is the earning capacity of any business.

If a business or firm takes loan from any bank or generate debt on which there will be some cost as we have to pay this amount by considering the rate of interest. If this interest rate is less than that earning capacity which a firm or business has then there will be financial leverage created and the profit of shareholders will get increased.

Return on Equity (ROE): When earning capacity will be on certain level and is more than cost of debt then the return of shareholders or equity holders get increases due to the increase in financial leverage. And return on equity will be more and more with a Capital Structure having higher percentage of financial leverage.

Leverage and ROE: This is the relationship of leverage that more is the leverage; more is the return on equity provided that the cost of debt is less than the overall earning capacity of the firm.

So, the firm, the management of that firm or business and also investors already knows this relationship. So, in both the cases when leverage increases, the shareholders increase their required rate of return.

Cost of Equity Capital: This is actually the required rate of return of the shareholder that shareholders want to earn how much amount of return or profit from any firm or business. This required rate of return which is the cost of equity capital changes with the change in the Capital Structure.

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Topic 206: Traditional Theory of Capital Structure-2

In previous lecture, we have started our discussion about Traditional Theory.

Traditional Theory of capital structure simply states that there exist an optimal capital structure and where the value of firm is maximum and the cost of overall capital is minimum.

How return on equity, the required rate of return and the leverage is related to the Traditional Theory?

Return on Equity (ROE):

Return on Equity (ROE) is the return of shareholders and this return of the shareholders is actually something which increases or decreases because of the leverage. When we take loan and invest in our business and earn more than the interest paid on that amount of loan then this excessive amount is included in the profit of equity holders and hence the return of equity holders actually increases.

So, the **Return on Equity (ROE)** is dependent on the earning power of the firm or the business.

Formula to calculate the Return on Equity (ROE):

Return on Equity-ROE = PAT/Equity

Where 'PAT' is profit after tax which is the final figure of profit which is calculated by deducting all the cost and expenses and financial cost and taxes out of the Revenue.

'Equity' is the total amount or the total value of investment of the shareholders in the company.

Cost of debt and Earning Power: The cost of debt and the earning power of a company should be in a certain relationship meaning that cost of debt should be less than the earning power of a company then only financial leverage will work and equity holders will increase their return due to that increase of financial leverage. We can say that all the theories of Capital Structure is dependent on this assumption that **cost of debt remains less than our overall earning power**.

General Assumptions of Theories:

There are some assumptions related to Traditional Theory of Capital Structure and we see that what are our basic assumptions on which this theory has been built up.

Assumptions are actually those things which we suppose that they will remained fixed at certain level or certain method or certain percentage or amount and they

will not be affected by any other factor. Some of these assumptions are realistic and practical but in some cases the assumptions are not that practical so, at times we have to remove those assumptions and see that what is the impact, if this assumption is not true.

- The **first assumption** is that the cost of debt should be less than the cost of equity or we can say that equity has a higher required rate of return than the debt.
- The **second assumption** is that we assume that there are no taxes, in the practical world there are corporate taxes and individual taxes meaning that we live in such a world where in any country we have to pay taxes on our income or profits. And if we are talking about such a world for applying capital structure theories where there are no taxes then it may be unrealistic assumption but by the way, this assumption is taking in the basic application of Traditional Theory or other Theories.
- And finally the **third assumption** is that the risk perception of investors does not change by the use of debt. So, it is said that people as investors, they do not consider that the risk perception or the risk profile of an organization has changed when the organization has borrowed more amount of debt or loan.

So, we can say that in the **Traditional Theory of Capital Structure**, the value of firm is considered to be maximum with a minimum cost but this is actually done with certain assumptions.

We can conclude our discussion that we can say that the value of firm in the Traditional Theory is dependent on the cost of capital and the cost of capital reduces with the increase in the leverage and up till certain point this cost reduces after which there is an increase and ultimately we reach to an **Optimal Capital Structure**.

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Topic 207: Traditional Theory of Capital Structure-3

The Traditional Theory of Capital Structure actually states that with the increase in the leverage, the shareholders increase their required rate of return.

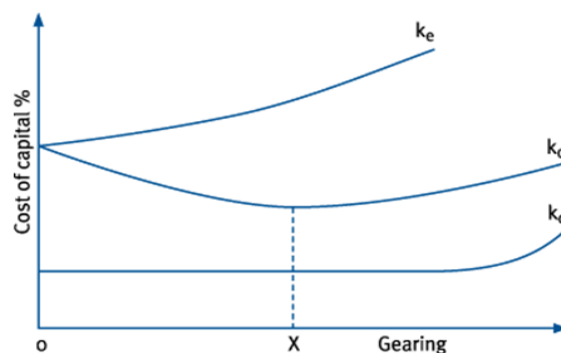
The **basic difference in both theories** is that **Traditional theory** states that when financial leverage increases their required rate of return but they don't increase to that extent that overall cost of capital remains as it is and instead of that they initially increase less their required rate of return.

Now, increasing the required rate of return a little and not in proportion with increase in the financial leverage results into the reduction of overall cost of capital.

And then we reach to a point where the overall cost of capital is minimum and from here we further increase the financial leverage of our business or take more debt then shareholders see that there is a lot of financial leverage and our return should get increased. In that case, they increase their required rate of return to such an extent that overall cost of capital instead of decreasing, it starts increasing.

Traditional Theory

- With the increase in leverage the cost of equity increases
- Initially the overall cost goes down
- Later overall cost goes up
- At the point where cost is minimum Value is maximum



Here ' k_o ' is the overall cost of capital and this is the cost which is used to determine the value of firm. Initially this graph is going down until it reaches to the point X. **On the x-axis,** we have got leverage or gearing that is another name for

leverage and on the y-axis, we have got the cost of capital. So, we can see that as the leverage is increasing, the cost of capital is decreasing and at the point X, it becomes minimum. This is the point where the value of the firm is maximum and this is a point of **Optimal Capital Structure**.

Now, this initial reduction is because of the increase in the required rate of return less than the overall increase in the financial leverage.

And later the required rate of return is increased with such a proportion that it by passes that it overtakes the percentage increase of leverage.

And that is the point of minimum value of 'k_o' or WACC (Weighted Average Cost of Capital) and that is a point of maximum value of firm.

Now, to conclude our discussion on **Traditional Theory of Capital Structure**, we can say that Traditional Theory is a theory of Relevance of Capital Structure whose basic point is that Capital Structure affect the value of our firm and other basic point is that the value of firm changes as the capital structure changes and there becomes such a point where our value of firm becomes maximum which we call **Optimal Capital Structure**.

So, every firm or business should always try to revise its capital structure again and again that it reaches at a level where its overall cost of capital become minimum and the value of firm will become maximum.

We can conclude our discussion that the value of Firm can be changed or increased or maximized or can be increased to the maximum level which is called the Optimal level. So, the capital structure is something which is Relevant meaning that capital structure is not such a thing which does not affect the value of firm. And every firm or business wants to increase or maximize the value of its firm and to increase the value of firm in addition to other things that the management could do; one thing that they should do is to alter and improve the capital structure.

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Topic 208: Net Operating Income Approach-1

In this lecture, we are shifting our discussion from the Traditional Theory or the Relevance Theory to the Irrelevance Theory.

The Irrelevance Theory is not just Net Operating Income Approach but there are other theories, this is one which is representative Irrelevance Theory of Capital Structure and we will discuss the other popular argument which is called the M & M argument in the other lectures.

The Theory

- Capital structure has no impact on the value of firm
- Overall cost of capital does not change with the change in capital structure
- There is no optimal capital structure
- Value of firm is based on earnings

Irrelevance Theory is strictly the opposite of Traditional Theory. For example, the Traditional Theory states that capital structure has a direct impact on the value of a firm. So, Net Operating Income Approach states that no, capital structure is not affecting the value of firm or capital structure has no impact on the value of firm.

That means that overall cost of capital does not change with the change in the capital structure meaning that basically the Traditional Theory states two things and these are; our cost of capital is less in the start by changing the capital structure and then starts to increase later and reached at a point where cost of capital becomes minimum and that point is called the Optimal Capital Structure.

But Net operating Income theory states that when we increase the financial leverage then there is no change in our overall cost of capital with this increase.

Why there is no change in the overall cost of capital by increasing the financial leverage?

The answer is that whenever a firm increases leverage and changes its capital structure, the investors or the shareholders knowing the impact of the change in the capital structure, they change or increase their required rate of return and they exactly increase their required rate of return to the extent that the overall weighted average cost of capital remains the same.

It means that the **difference between Traditional Theory and Irrelevance Theory** is that **Traditional theory states** that the required rate of return increases with the increase of leverage but Traditional theory says that at the start, the required rate of return of shareholders increase less and later on it increases more whereas **Net Operating Income approach** says that no, the required rate of return of shareholders increases with the proportion as with the increase of leverage that there has not been changes in the overall cost of capital.

So, if the overall cost of capital has not been changed then **there is no Optimal Capital Structure** or in other words, every capital structure is an Optimal Capital Structure that means that instead of trying to alter the capital structure, alter the leverage and trying to increase the value of a firm, the management should try to do something else to increase the value of the firm.

It means that when **Net Operating Income approach** negate the affect or impact of capital structure on the value then next question is created that **value of firm increase or changes from which thing** and the argument is simply that **it is the income which increases the value of a firm**. So, if the firm wants to increase its value, it has to increase its earnings or profits.

So, we can conclude our discussion that **Net Operating Income approach** is the total contrast to the Traditional Theory and this approach states that there does not exist any Optimal Capital Structure meaning that shareholders proportionally according to the leverage, continue to adjust their required rate of return and at every level of leverage and every level of capital structure, overall cost of capital remains same and there does not happens any change in the value of our firm. There may change happens that **Net Operating Income approach states** that it is

the earning to increase the value that if we need to increase the earnings of the business.

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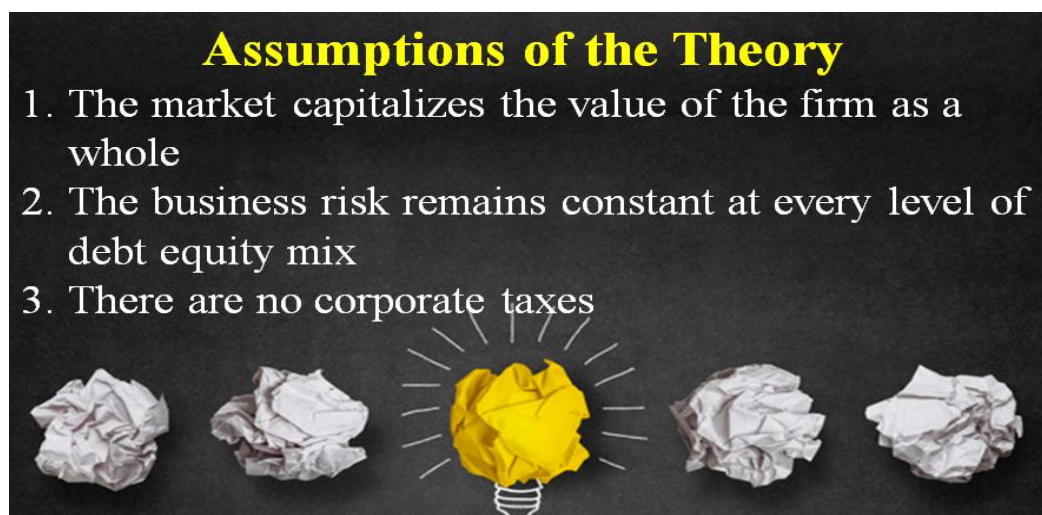
Topic 209: Net Operating Income Approach-2

We are talking about the Capital Structure theories and started our discussion from Traditional capital theory or Relevance theory and now we are continuing our discussion of Net Operating Income approach of capital structure or Irrelevance theory.

Assumptions of Net Operating Income approach or Irrelevance Theory:

There are certain assumptions of Net operating income approach as these theories actually do not work in isolation. Whenever we have to apply these theories in practical world then there may be many restrictions or limitations that we have to face which we write in the form of assumptions with that theory.

It means that if these theories hold these assumptions then the argument in these theories will be validated.



The first assumption is that the market capitalizes the value of the firm as a whole. It means that we have an overall capitalization rate through which we can calculate the value of firm.

The business risk remains constant at every level of debt and equity mix. This is not a very realistic assumption because when we increase the leverage and then there has been changes in the risk profile but in this theory, this assumption is taken that the risk level does not change.

The third assumption is same as in the Traditional Theory that there are no corporate taxes. And this is again not a very realistic assumption, having said this; these theories can then also be discussed or explained by not assuming that there are no corporate taxes. It means that we can see by applying these theories that if there are corporate taxes then how this theory will be applicable.

There is a very interesting argument of Net Operating Income theory and that argument is about the value of a π . This theory says that if the value of firm can be changed by changing the proportion of equity and debt then if π (circular object) can be cut into three pieces instead of two pieces in which one piece may be small and other part may be large or vice versa then there should be no changes in the value of π .

Value of Pie Remains the Same



- Value is affected by profit not the capital structure
- No matter in how many pieces the pie is cut the total value does not change

And Net Operating Income approach says that it does not matter that in how many pieces you cut a π whether we make 2 pieces of a π or we may 4 pieces of a π or 8 pieces of a π or more or less than that then we sum up or when add all those pieces, the total value will be the same and that is the total value of the π .

That is 360° , like if we make 4 pieces then each piece will be of 90° ($360/4 = 90$) and we make 2 pieces then each pieces will be of 180° ($360/2 = 180$) or more or less than that but we sum them up then their total value always be same as that is 360° .

The argument is then applied on the value of firm that we say that a firm or a business is a π and capital structure debt and equity are the portions of that value of the firm. So, if we increase or decrease debt and to increase or decrease equity then the proportion would change but the sum of the two will always be the same.

- So, this results into a conclusion that the value of a firm is affected by the profit and not by the capital structure. Because it only changes the proportion, it does not add the value so, we have to earn profit to increase the value and this profit is actually the increase in the value.
So, the argument is that try not to cut the π in different pieces in different proportions to increase the value but try to increase the profits or earnings if we want to increase the value of a firm. So, the capital structure is Irrelevant.
- We can summarize this argument by saying that the Net Operating Income Approach states that no matter in how many pieces a pie (π) is cut the total value does not change. That means if it is applied to the capital structure we can say that no matter in what proportion we divide the overall capital of a firm into debt and equity, the total value of that capital and the total value of the firm will not change.

We can conclude our discussion on the Net Operating Income Approach, we say that this is an Irrelevance Approach and it says that capital has nothing to do with the value and that is the value of firm is not dependent on the capital structure.

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Topic 210: Net Operating Income Approach-3

In Net Operating Income Approach, we have discussed that the capital structure does not affect the value of a firm and the arguments which are given by the proponents of Net Operating Income Approach are different from that of the Traditional Approach.

The Argument

1. Impact of leverage on cost of equity
2. Rate of change of cost of equity
3. Impact on overall cost of capital
4. Impact on value of firm



If we say that both approaches (Traditional Theory and Net Operating Income Approach) are contrast to each other but in one way the arguments are somewhat similar meaning that what is the impact of leverage on the cost of capital and this is one thing in two theories are somewhat similar that the leverage has an impact the cost of equity.

When leverage will increase then as a result cost of equity will also get increase. So, the argument is that the cost of equity increases but how much is the increase in the cost of equity. This is the slight difference between the two theories (Traditional theory and Net Operating Income theory).

The Traditional Theory states that with the increase in the leverage, initially the cost of equity increases to the extent that overall cost of capital reduces and this keeps on reducing until a point which is the saturation point and after which the overall cost of capital starts increasing because the shareholders enhance or

increase their required rate of return to a great extent increasing the overall cost of capital.

The **Net Operating Income Approach** states that yes, when the leverage in increased, when the debt proportion is increased then this certainly results into increase in the required rate of return of the shareholder or the cost of equity but this increase in the cost of equity is exactly the same which keeps the overall cost of the capital same.

How this decision can be made that how much required rate of return the shareholder will increase and this is something which is related to the behavior in finance or the behavioral aspect of the shareholders. We have to see to test this thing by doing experiments that if we change the leverage or capital structure of any firm or company then what will be the changes in the overall cost of capital by this change.

But when we actually try to apply or implement this theory; there are many other factors which actually affect the cost of capital or the market price of the debt and equity enhance the value of the firm. And these **external factors** which may be political factors, economic factors, social factors, international changes in the world and all these factors has somehow an effect or impact on a particular business which is being carried out in a particular economy or a country.

So, that means it is impossible to say that keeping everything else constant, we just see the effect of leverage. At one time, the value of firm or the market price or the impact on the required rate of return is coming of many factors so, it is not possible for us to conduct an experiment where we keep the economy, the politics, society, the inflow of cash from other countries; all these things are constant and see that the impact of only the capital structure changes.

Therefore we say that both these are like suppositions or like the predictions of the behavior of shareholders that one theory says that the required rate of return of shareholders will be less in the start and will be more at the end and whereas other theory says that no, it increases proportionally therefore there has no change happens in overall cost of capital.

We can also have some calculations where we can consider two companies which are having some debt and company not having debt and then see that whether this theory A works or theory B works on that.

Now, the impact of overall cost on the capital is something that is key contention in these two theories. That the overall cost changes the value of the firm, the overall cost of capital actually changes with different rates and that is said by the proponents of these theories.

Our concluding arguments about the Traditional Theory and Net Operating Income approach that these two theories talk about the behavioral aspects of the investors.

That how do they behave once they invest in a certain company when they buy the stocks of a particular company and how do they react and how do they behave when the company changes its capital structure.

We may have to control many things in order to do this observation as we have discussed and we name them Control variables in the research meaning that if we may control these things then we can see that what will be the behavioral aspect of people.

The behavioral aspects of the shareholders or investors changes will the time, changes from place to the place and there are several other factors which influence the behavior of the investors or shareholders.

So, we can conclude our discussion that the two theories will certain set of assumptions can be proved with certain values or numerical values but if remove those assumptions, these theories are not proved as true or false.

Module 11: Capital Structure

Topic 211: Arbitrage Theory: Miller and Modigliani-1

We have already gone through the discussion of Relevance and Irrelevance of Capital Structure.

We have discussed that there are two theories which exist; the Traditional Theory which states that Capital Structure is Relevant, it has an impact on the value of a firm. Whereas we had a discussion on Net Operating Income Approach which states that the Capital Structure does not affect the value of a firm.

In this lecture, we are going to discuss another theory or another approach or another supportive argument for the theory which is called the Irrelevance Theory of Capital Structure.

The Irrelevance Theory or argument was built by **Miller and Modigliani** and that is way it is called M & M theory.

Irrelevance Theory of Miller and Modigliani argument:

Irrelevance Theory

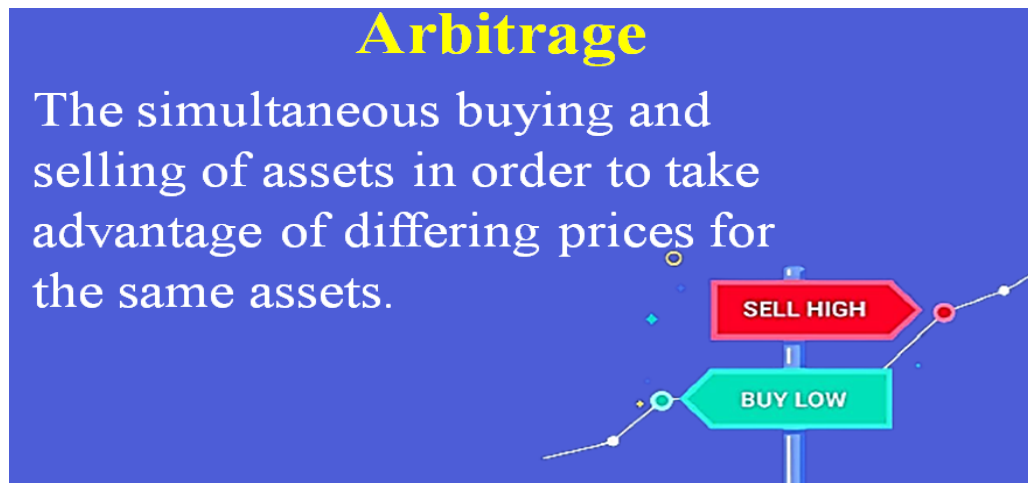
1. Capital structure is irrelevant w.r.t. value of a firm
2. If relevance theory (Traditional Theory) is accepted, Arbitrage will take place.



The first one is same as that of the Net Operating Income approach and that is; Capital Structure is Irrelevant with respect to the value of a firm. So, if a firm wants to increase its value that cannot be done by altering the capital structure or by changing the leverage.

If the Irrelevance theory is applied in a way that the concept of Arbitrage was introduced by M & M and they said that if we for a moment assume that the

Traditional Theory or Relevance Theory is correct or accepted then in the market Arbitrage will take place.



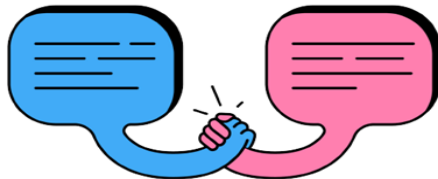
Arbitrage is actually a process in which we buy an asset which is cheaper and sell an asset which is expensive, if the two assets are identical. It means that the basic argument is that it is not possible that two assets which are identical (I.e., shares and stocks) but even if apply to any assets; **Arbitrage** states that it is not possible to have two assets which are identical and they are selling at two different prices in the market.

It is because that everybody would like to sell the expensive asset and buy back the asset which is cheaper and in this way we can save our some money and similar asset will be there with us.

M & M (Miller and Modigliani) Argument:

M&M Argument

1. Two same assets (stocks) cannot have different prices
OR arbitrage will take place.
2. The prices of identical firms' (other than capital structure) stocks cannot be different.



They said that the Relevance of dividend saying that the two firms which are same and every context other than the capital structure cannot have different prices in the market or different value in the market. It is not possible and if it will be done then it will create Arbitrage.

It means that when we say that investors behave rationally where rational behavior means that every investor wants to earn more profit and where there is opportunity then he avail that opportunity to enhance his/her profit.

So, M & M argument is as:

- Two assets which are same or the two stocks or two companies or the firms cannot be different in terms of their prices in the market meaning that if those two stocks or companies are same then their prices cannot be different otherwise Arbitrage takes place.
- Secondly, the prices of identical firms (firms which are identical in every aspect other than the capital structure) cannot be different.

We can conclude our discussion that M & M (**Miller and Modigliani**) has introduced the process of Arbitrage with reference to the capital structure. And they said that if we assume that two same companies which are similar in every aspect but their capital structure (there is difference between their debt and equity ratio as somewhere leverage is more or somewhere leverage is less) is different then their market prices cannot be different and if they will be different then Arbitrage takes places meaning that people will sell their shares or stocks to

expensive companies and buy the shares or stocks of cheap companies and in this way they will save some money.

Module 11: Capital Structure

Topic 212: Arbitrage Theory: Miller and Modigliani-2

In previous lectures, we have discussed that the Irrelevance Theory is supported by two gentlemen Miller and Modigliani who have contributed a lot in finance and their M & M arguments or theories are there in finance for the noble laureates the Franco Modigliani and Merton Miller.

M&M Argument

1. Personal leverage
2. Cost of debt is same for firm and individuals
3. Investors shift to the other firm with same return and value and yet save some money.



Franco Modigliani



Merton Miller

The argument that they gave actually was that if an investor gets an idea that the similar stocks are trading at different prices, they will get into Arbitrage.

How would they get into the Arbitrage process?

Suppose, if there is a company whose value in the market is more and there is also another company whose value in the market is less then they will shift their investment from the expensive company to the cheaper company and in this way, they will save some money.

- The problem or the discussion here is that if we are moving from one firm to the other that other firm would also have some leverage and if we are selling

our stocks as an investor to get into Arbitrage, our stocks' value will not be the same as that of the other firm because of the leverage.

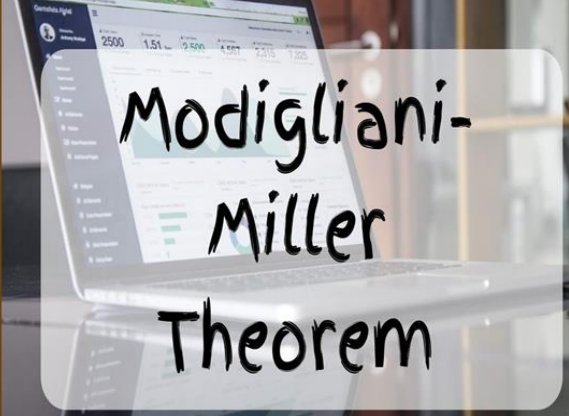
1. They said or they assumed that personal leverage is possible meaning that if a company or a firm take loan for its business and similarly an individual can also take loan and invest it.
2. The second argument or assumption is that the cost of debt is same for the individual as that of the firm. It means that if a firm can get loan on 10% per annum then individual can also get loan on that same rate. And again this is not a realistic assumption because of the risk profile of an individual which is very high as compare to the firm. But this is the second part of their argument. To create an Arbitrage situation, this is what an investor has to do.
3. The investor after taking the loan will invest or shift the money in the firm which is cheaper and in this way save some money having the same return.

It means that when he will shift from one firm to the other and because of this shifting he will have get same percentage holding and will have get the same value of investment but some debt will be created there because he is holding that percentage by borrowing. And due to this borrowing, he has to pay some interest about which we have said that its cost be same that this company or firm has to pay. In this way, he will create its Arbitrage and as a result of that Arbitrage, at the end he will be able to save some money out of that.

M & M Arguments:

M&M Argument

1. Percentage holding will not change
2. Return will not change
3. There will be cash saving



Modigliani-
Miller
Theorem

1. In this theory of **Miller and Modigliani's** Irrelevance about Capital Structure, the important thing which may be a concern that if someone say that our percentage holding will change meaning that we have 5% holding in a company and we shift it to the other company then if our percentage holding becomes less then our influence can get reduced or our involvement in the decision making may get reduced and other similar factors. By creating an individual leverage that is borrowing the money from financial institution at the same cost, investor will be able to keep the percentage holding same or percentage holding will not change while shifting from one firm to the other.
2. The second thing is that the net return, the return in the other firm will be little higher but once the cost of debt will be subtracted and percentage is calculated in Arbitrage, shifting from one firm to the other still give the investor same rate of return or the return will not change but the incentive or the attraction.
3. When we say that there is no difference in the percentage holding and its rate of return will not be changed then **why there will be Arbitrage**. It will be because of that there will be some cash saving by investor through all this process which will be the return or additional benefit that is because of the Arbitrage process.

We can conclude our discussion that Miller and Modigliani has suggested such process of Arbitrage that is based on that we have to create self-made leverage. This self-made leverage is with such assumption that there will be no changes in the borrowing rate and borrowing can be done with the same frequency whenever it has to shift then will get that money.

If it is admitted then the argument of **Miller and Modigliani** is that through the Arbitrage, there will not be change in the holding percentage which is one concern and its rate of return will also not be changed which is another most important concern but the cash saving will certainly induced all the investors to get into the process of Arbitrage.

Module 11: Capital Structure

Topic 213: Arbitrage Theory: Miller and Modigliani-3

In our discussion of Irrelevance Theory and **Miller and Modigliani** argument, we have discussed that why and how Arbitrage will take place and how investors will be able to the same return by shifting from an expensive company or a firm to a cheaper company or a firm and keeping their percentage holding the same but saving some amount of cash.

If this will as it has been said then there will be Arbitrage and **what will the result when this Arbitrage exists?**

This is an important thing that we need to understand that **Arbitrage** will induce for selling the shares or stocks of the expensive company that will create an excessive supply of that expensive company. When all the people or investors are behaving rationally and they will start to sell the shares of expensive company in order to buy the shares of cheaper company then it will create an **excessive supply**.

On the other hand, all these people who are getting into this Arbitrage will try to get into the cheaper company meaning that they will try to buy that company whose price is less and that will create a demand.

So, this means that excessive supply of one stock and then the excessive demand of the other. When there will be excessive supply of expensive company then there is a **basic principle of a stock market** is that **excessive supply results into reduction in the price** and the **excessive demand of the cheaper company creates an increase in the price**. This process of Arbitrage will happen too quickly as **Miller and Modigliani** said that in few moments, the price of both shares or stocks will become same and that will be the moment when Arbitrage process will stop.

General Assumptions of M & M argument:

General Assumptions

1. Firms can be identical
2. External factors do not affect
3. No corporate taxes
4. Perfect market
5. Rational behavior of the investor
6. Identical risk characteristics
7. All earnings are paid as dividend



1. The first assumption is that firms can be identical meaning the term firms are identical means identical in every aspect other than the capital structure. So, we actually know that two firms cannot be identical even if we try to consider two firms of somewhat similar or same capital amount or other attributes but two firms in the real world cannot be identical.
2. The second assumption is that external factors do not affect. Whenever we talk about any stock or commodity market or in every market, we know that there are many external factors which do affect the market and the prices go up and down because of the influence of all those external factors. So, this assumption again is not realistic and this brings us into a market or world which is based on another assumption which is unrealistic.
3. As we have discussed earlier in other two theories (Traditional theory or Relevance theory and Net Operating income theory or Irrelevance theory) of capital structure that there are no corporate taxes. But practically we do have corporate taxes.
4. There is another assumption is Arbitrage process is that markets are perfect. The Perfect Market is such a market in which there are many assumptions; product will be identical, buying or selling will be rationally, there will be no barrier on entry or exit etc. this is that market which exist theoretically but it is not possible to find a market which in its real sense a perfect market.
5. Every individual always behave Rationally. Unfortunately it is not again very true that investors generally do behave rationally but many investors

follow the herd, we have seen very commonly that herd behavior has been seen and at times emotional decisions are made or speculative buying and selling is done. So that Rationality is also an argument which may not be 100% true for the investment markets.

6. The firms have identical risk characteristics which in practical world is again very difficult to achieve.
7. The last assumption is that all earnings are paid as dividend which means that there is no growth in the company.

We can say this conclusively by seeing all these assumptions that **Miller and Modigliani** argument is valid with so many things are kept constant. It means that these many assumptions are considered to be true and these many variables have been controlled which is in practical world is impossible; only then this Arbitrage process in this theory will be workable.

Module 11: Capital Structure

Topic 214: Market Imperfections and Issues

We have discussed in detail that what is the argument of M & M and how it is applied but then we talked about several assumptions which are there as a part of the theory of M & M.

Now we are considering some of them in detail;

The first one is that Market imperfections and other issues related to that.

Perfect Market

- A theoretical market
- No product differentiation
- Large buyers and sellers
- No entry or exit barriers
- Perfect information
- Rationality
- No monopoly or other manipulations



Perfect Market:

A theoretical Market: A Perfect Market is a theoretical market; the market which actually does not exist because of so many reasons.

No Product Differentiation: The Perfect Market assumes that there is no product differentiation meaning that if we are talking about stocks, there are stocks which are identical and we have said that stocks or companies or firms cannot be identical.

Large buyers and sellers: There is a very large number of buyers and sellers. Now, if we see the practical markets then we see that the number of buyers and sellers at times is large but at times it is not large. For some stocks, we would see that there may not be any buyer or any seller.

No entry or exit barriers: Likewise, it is said that there are no entry and exit barriers. Anyone who wants to get into the market to buy or sell but when we talk about actual markets, these are regulated markets and in the regulated markets there are some barriers.

Perfect information: There is perfect information which is available to everyone meaning that it is assumed that every investor who will trade in that market will have all the correct information and on the basis of this information investor will rationally (rationality simply means that to make decision in order to maximize own profit) maximize his profit.

No monopoly or other manipulations: It has been assumed that there are no manipulated factors or any monopoly in the market. The **monopoly** is the concept of economic in which we say that one buyer or seller becomes larger enough that he may influence the whole market and can fluctuate the prices by his own self. So, this is another assumption of Perfect Market.

Impact of Market Imperfections:

Having said this this that markets cannot be perfect then what happens if there are imperfections.

Impact of Market Imperfections

- All real market are imperfect
- Manipulation
- No possibility of Arbitrage
- M&M theory does not hold good in imperfect markets



All real markets are imperfect: If all real markets are imperfect then all these possibilities or assumptions are not valid in the practical market.

Manipulation: There is practical manipulation; we find that bigger investors or institutional investors or mutual funds they are in a position to somewhat affect the prices in the market.

No possibility of Arbitrage: If this is something that happens because Arbitrage is only possible if these forces are not there or if the markets are perfect. So, we can say that there is no possibility of Arbitrage and the theory of M & M would not hold good in that situation.

M & M theory does not hold good in imperfect markets: We may say that M & M theory is application with several assumptions; one of which very important is perfect market and if perfect market does not exist then this theory does not hold good.

Issues in M & M Argument:



Bankruptcy cost: We say that as the financial leverage increases, the risk level of firm also increases with it and increase in the risk level also increases the chances of bankruptcy. And if a company gets bankrupted then everything is lost so, there is a cost associated to the bankruptcy which also should be considered in a theory but M & M does not consider it.

Agency cost: There is another theory which is called the Agency theory. The **Agency theory** states that investors do not directly control a business or a firm or company; the investors indirectly through the manager or directors of the company have an influence on the company and their decision making.

Because of this Agency issue, there is an associated cost of that but M & M does not also consider the cost of Agency which again creates a possibility that the M & M argument will not be valid if there will be agency cost.

Leverage increases efficiency: There is another interesting argument which say that when financial leverage starts to increase due to which risk get increases and due the increase of risk, there happens naturally an increase of efficiency in the management of firm or business because they start working on their toes or they try to put an extra effort so that the high risk profile can be controlled and kept within the limits.

So, wherever there will be higher efficiencies then there will be some impact of that.

Institutional restrictions: Another issue is that there are some institutional restrictions. The institutional restrictions means that buying or selling the shares, institutions have their own policies what to sell, how much to sell, when to buy, when not to buy, which companies should be considered for investment or not. But this factor is also not addressed by M & M theory.

Transactional Costs: We can say that with these issues and this cost of buying or selling that includes the brokerage charges; when we buy or sell shares then we have to pay some commission which again has different rates but also influences the decision of buying or selling or changes the cost of buying or selling.

We do say that M & M does not consider for this transactional cost.

We can conclude our discussion that “**in an imperfect market capital structure may affect the value of a firm.**”

And our basic argument was that there has not been happen any difference in the value of the firm from capital structure and that argument holds only when our markets are perfect. And we have seen due to many issues and reasons that markets are not perfect. So, as a conclusion we can say that M & M theory is only applicable in perfect markets and not in imperfect market.

Module 11: Capital Structure

Topic 215: Income Tax Considerations

In previous lecture, we have talked about the Capital Structures Theories and we have seen that Relevance and Irrelevance and also a long set of assumptions which are there when one theory or the other theory is presented.

One of the very basic assumptions was that we live in a world where there are no taxes and that was an unrealistic assumption because we actually live in a world where there are taxes.

There are two types of taxes that we need to understand as follows:

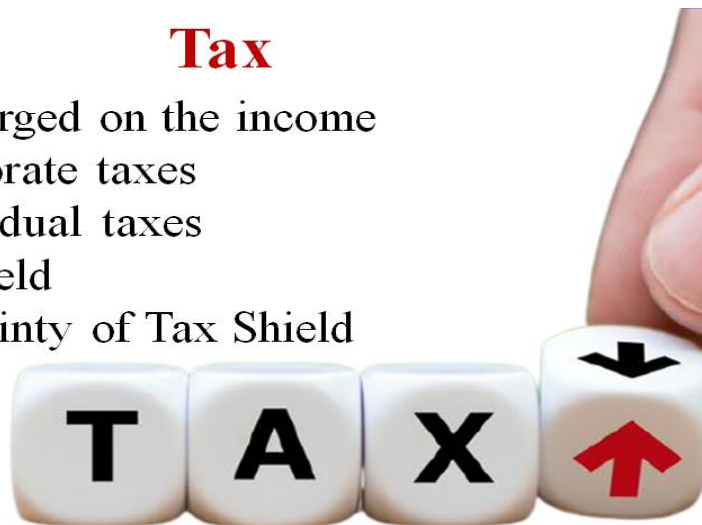
One type of taxes is those which we name Indirect Taxes in the today's world meaning those taxes which are applied on expense, if we purchase something or do any expense then we have to pay some tax on this.

So, we call this tax as Indirect Tax i.e., sales tax, excise duty etc.

Other type of tax is that we have to pay on our profit or earning and this is proportional to our earnings meaning that as our earning will increase then our tax will also get increased.

Tax

- Tax charged on the income
 - Corporate taxes
 - Individual taxes
- Tax Shield
- Uncertainty of Tax Shield



When we talk about the implication of taxes in the theories then we are actually referring to the Direct Tax that is Income Tax so, the tax that is charged on the income is the issue of the discussion here.

The income tax is applied in two different ways; for **Individuals** the tax rates and tax applications is quite different as compare to the **Corporate Taxes**. The companies, the businesses pay taxes at the different rates and in a different manner.

So, if we say that there are taxes then these taxes will not be the same for Individuals and for the Companies. If a company is trying to get into Arbitrage and buying & selling from one point to the other point, the tax consideration will affect that decision and that theory will not hold good. Other thing that we say is that our theory is not valid due the rates and the applications of Individual taxes and Corporate taxes.

Tax Shield:

Another very important concept is **Tax Shield**. When debt introduces in any firm or the component of debt is more then that company pays more interest on that debt and interest is admissible expense for the calculation of tax. It means that if company A is paying Rs.100 interest and company B is paying Rs.200 interest then company B has to pay less tax due to the payment of more interest as compare to the company A. So, this is called the concept of Tax Shield.

If the leverage is increasing or if the amount of debt is increasing that means that more interest will be paid and incurred and as a result the amount of profit will be reduced and hence the amount of tax will be reduced. So, if a firm has a higher leverage or have more debt then there will be more interest also and due to more interest, the profit will be less due to which amount of tax will be reduced. And other company will not match the amount of return with it which is the basic thing of M & M argument that after Arbitrage, the return of both firms for the investor becomes same.

So, **in Arbitrage** the return of the investor is shifting from one company to the other company, if there is income tax because of the Tax Shield will not be the same.

Uncertainty of Tax Shield:

Another important element with reference to income tax and the tax shield which is created that this tax shield is also uncertain and uncertain from the view point that tax rates are not fixed forever meaning that it changes with the passage of time.

We can conclude our discussion that once a company is exposed to the taxation which is the corporate taxes and the individual taxes are exposed to individual taxation and these tax rates are fluctuating, the tax shields are different. And there are uncertainties of tax shield also and M & M theory does not hold good.

So, again this assumption is very important to prove the process of Arbitrage and in an income tax world, M & M theory does not hold good.

Module 11: Capital Structure

Topic 216: Value of a Firm in the Presence of Income Tax

In the last lecture, we talked about the value of firm in the presence of corporate and individual tax. We have said that there will be corporate and individual taxes available which we named as Income Tax or Direct Tax so, in that case the value of both firms (one firm who is levered or has debt and other firm which is unlevered or has no debt) will not be equal.

Value of a Firm in a Taxable World

Value of levered firm is higher in a taxable world.

Value of a Levered (who has debt) Firm = Value of an Unlevered (who has no debt) Firm + Tax Shield Benefit

Value of a Levered Firm = 500,000 + 150,000 = Rs. 650,000

As Value of levered firm (here it is 650,000) is higher in a taxable world as compare to the unlevered firm (here it is 500,000) but again this statement is against the argument of M & M argument or Relevance theory.

Value of Tax Shield

- Tax Shield Benefit = Debt x Tax Rate
- Tax Shield Benefit = 500,000 x 30% = 500,000 x 0.3 = Rs. 150,000

Or

- Tax Shield Benefit = Tax Saving/Discount Rate
- Tax Shield Benefit = 22,500/15% = 22,500/0.15 = Rs. 150,000